

Task 5

Pseudocode

processed $\leftarrow 0$

standard-vehicle, large-vehicle $\leftarrow 0$

accept $\leftarrow 0$

reject $\leftarrow 0$

ZoneA-cap $\leftarrow 20$

zoneB-cap $\leftarrow 40$

zoneC-cap $\leftarrow 15$

OUTPUT "~~Enter~~ Enter total vehicles"

INPUT total-vehicles

FOR $x \leftarrow 1$ TO total-vehicles:

OUTPUT "Enter Vehicle Type": (C)ar, (B)ike, (V)an

INPUT vehicle-type

OUTPUT "Enter user category: (F)aculty, (S)tudent, (G)uest

INPUT user_cat

OUTPUT "Valid parking permit?: (Y)es or (N)o"

INPUT permit

OUTPUT "Emergency Vehicle? (Y)es or (N)o"

INPUT emergency

vehicle-type !=

AND $\downarrow$ AND $\downarrow$

WHILE vehicle-type != 'C' OR 'B' OR 'V':

OUTPUT "Enter valid vehicle type"

INPUT vehicle-type

ENDWHILE

user_cat !=

user_cat !=

AND

AND

```
WHILE user_cat != 'F' OR 'S' OR 'G'  
    OUTPUT "Enter valid user category"  
    INPUT user_cat  
ENDWHILE
```

AND permit !=

```
WHILE permit != 'Y' OR 'N'  
    OUTPUT "Enter valid parking permit status"  
    INPUT permit  
ENDWHILE
```

AND emergency !=

```
WHILE emergency != 'Y' OR 'N'  
    OUTPUT "Enter valid emergency"  
    INPUT emergency  
ENDWHILE
```

IF permit = 'Y' ↓ section 2 start

IF user_cat = 'F'

IF zoneA_cap > 0:

IF vehicle_type = 'B' OR 'C'

success ← success + 1

zoneA_cap ← zoneA_cap - 1

OUTPUT "Park in Zone A" → OUTPUT "Remaining Capacity:" zoneA_cap

standard_vehicle ← standard_vehicle + 1

processed ← processed + 1

ELSE:

IF zoneA_cap > 1:

success ← success + 1

zoneA_cap ← zoneA_cap - 2

OUTPUT "Remaining capacity", zone2-cap
OUTPUT "Park in zone A"

large_vehicle $\leftarrow$ large_vehicle + 1

processed $\leftarrow$ processed + 1

ELSE:

OUTPUT "No available space"

reject $\leftarrow$ reject + 1

processed $\leftarrow$ processed + 1

ENDIF

ENDIF

ELSE:

OUTPUT "No available space"

reject $\leftarrow$ reject + 1

ENDIF

ELSE IF user_cat = 'S':

IF vehicle-type = 'B' OR 'C'

IF zoneB-cap > 0:

success $\leftarrow$ success + 1

zoneB-cap $\leftarrow$ zoneB-cap - 1

OUTPUT "Park in zone B"

OUTPUT "Remaining space", zoneB-cap

standard_vehicle $\leftarrow$ standard_vehicle + 1

processed $\leftarrow$ processed + 1

ELSE:

OUTPUT "No available space"

reject $\leftarrow$ reject + 1

processed $\leftarrow$ processed + 1

ENDIF

ELSE:

IF zoneC_cap > 1:

Success ← success + 1

zoneC_cap ← zoneC_cap - 2

OUTPUT "Park in zone C"

processed ← processed + 1

OUTPUT "Remaining space:", zoneC_cap

ELSE:

reject ← reject + 1

OUTPUT "No available space" ← processed ← processed + 1

ENDIF

ENDIF

ELSE IF user_cat = 'V':

IF zoneC_cap > 0:

IF vehicle_type = 'B' OR 'C':

Success ← success + 1

zoneC_cap ← zoneC_cap - 1

OUTPUT "Park in zone C"

OUTPUT "Remaining space:", zoneC_cap

standard_vehicle ← standard_vehicle + 1

processed ← processed + 1

ELSE:

IF zoneC_cap > 1:

Success ← success + 1

OUTPUT "Park in zone C"

zoneC_cap ← zoneC_cap - 2

OUTPUT "Remaining space:", zoneC_cap

large_vehicle ← large_vehicle + 1

Processed ← processed + 1

ELSE:

OUTPUT "No available space"

reject $\leftarrow$ reject + 1 $\leftarrow$ processed $\leftarrow$ processed + 1

ENDIF

ELSE:

OUTPUT "No average space"

reject $\leftarrow$ reject + 1 $\leftarrow$ processed $\leftarrow$ processed + 1

ENDIF]

End of section 2

ENDIF

ELSE:

IF emergency = 'Y':

(Repeat Section 2)

ELSE:

OUTPUT "No permit"

ENDIF

ENDIF

reject $\leftarrow$ reject + 1

processed $\leftarrow$ processed + 1

NEXT x

OUTPUT "Summary:"

OUTPUT "Total number of vehicles:", processed

OUTPUT "Total accepted vehicles", accept success

OUTPUT "Total rejected vehicles", reject

OUTPUT "Total number of cars and bikes successfully parked:", standard-vehicle

OUTPUT "Total number of vans successfully parked", large-vehicle

OUTPUT "Remaining capacity of zone A:", zoneA_cap

OUTPUT "Remaining capacity of zone B:", zoneB_cap

Zone A (Students) $\leftarrow 40$
Zone C (Visitors) $\leftarrow 15$

G, B, V
vehicle-type
user-cat
valid-permit

Task 5

~~Zone A $\leftarrow 20$~~

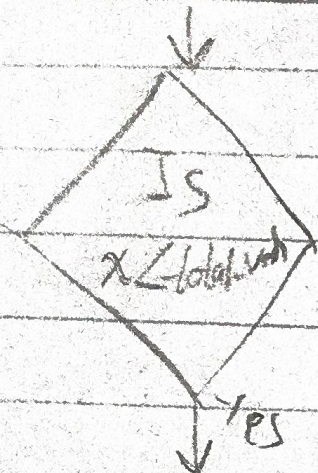
~~FOR~~

OUTPUT "Remaining capacity of zone C:", zoneC-cap
IF zoneA-cap $<$ zoneB-cap AND zoneA-cap $<$ zoneC-cap:
 OUTPUT "Zone A had highest occupancy"
ELSE IF zoneB-cap $<$ zoneA-cap AND zoneB-cap $<$ zoneC-cap:
 OUTPUT "Zone B had highest occupancy"
ELSE IF zoneC-cap $<$ zoneA-cap AND zoneC-cap $<$ zoneB-cap:
 OUTPUT "Zone C had highest occupancy"
ELSE IF ~~zoneA-cap~~ = 0 AND ~~zoneB-cap~~ = 0 AND
 zoneC-cap = 0:
 OUTPUT "Whole parking full"

Start

Zone A cap $\leftarrow$ 20
Zone B cap $\leftarrow$ 40
Zone C cap $\leftarrow$ 99

INPUT total vehicles



OUTPUT "Enter total vehicles"

INPUT "total vehicles"

OUTPUT "Enter user category"

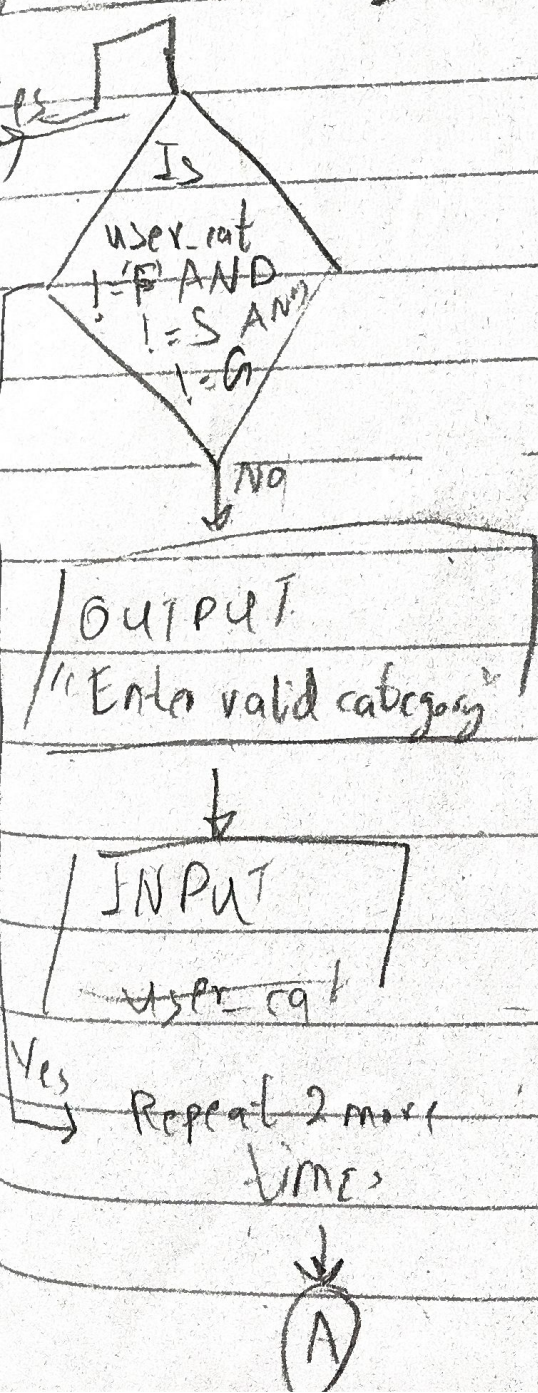
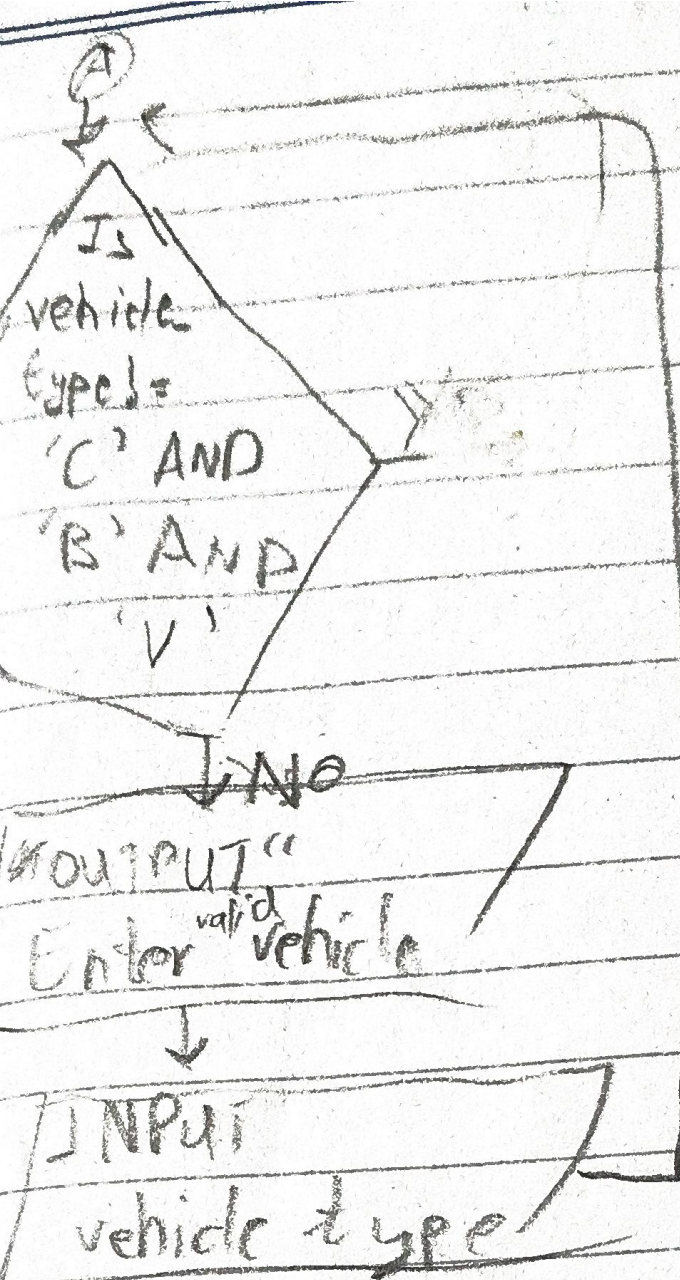
INPUT user cat

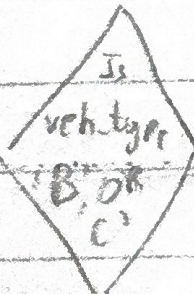
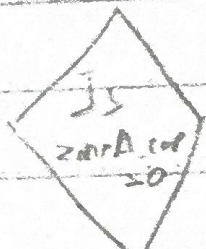
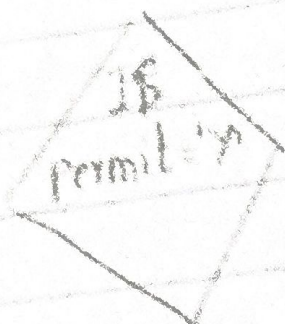
OUTPUT "Valid Parking Permit?"

INPUT permit

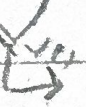
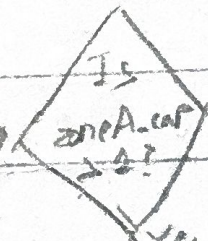
OUTPUT "Emergency Vehicle?"

INPUT emergency

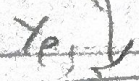




No



success ← success + 2



success ← success + 1

zoneA_cap ← zoneA_cap - 1



OUTPUT 'Park in zone A'

OUTPUT 'Remaining car' zoneA_cap



standard_vehicle ← standard_vehicle + 1
process ← process + 1

PAC:

Given data

zoneA_cap
zoneB_cap
zoneC_cap
~~processed~~
~~standard_vehicle~~
~~large_vehicle~~
~~accept~~
~~reject~~
total_vehicles
user_cat
permit
emergency

required results

processed
standard_vehicle
large_vehicle
accept
reject

Processing required

FOR $x \leftarrow 1$ TO total_vehicles..
-WH
-WHILE vehicle_type != C..
-WHILE user_cat != 'F'..
-WHILE permit != 'Y'..
-WHILE emergency != 'Y'..
IF permit = 'Y'
 IF user_cat = 'F'
 IF zoneA_cap > 0
 IF vehicle_type = 'B' OR 'C'

Alternative solutions

Set outputs as constants

IPO

INPUTS	PROCESSING	Module reference	Output
total vehicle	-success ← success + 1	Assignment addition	processed
vehicle type	-zone A_cap ← zone A_cap + 1	Assignment addition	standard vehicle
user category	-standard vehicle ← standard vehicle + 1	Assignment addition	large vehicle
permit			accept
emergency	-processed ← processed + 1	Assignment addition	reject
	-large vehicle ← large vehicle + 1	Assignment addition	
	-zone A_cap ← zone A_cap - 1	Assignment addition	
	-IF vehicle type = 'B' OR 'C'	condition	
	-IF zone A_cap > 0	condition	
	-IF zone B_cap > 0	condition	
	-IF user_cat = 'V'	condition	
	-IF zone C_cap > 0	condition	
	-zone C_cap ← zone C_cap - 1	condition	
	-IF emergency = 'Y'	condition	
	-OUTPUT "Enter vehicle type"	Output	